

Software Development, Maintenance & Operations Project: ML-KEM Legacy Modernization with LLM Assistance

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1 Project overview

In this group project, you will modernize a small legacy edge–cloud software system for the transition to post-quantum cryptography (PQC). The project combines software development, maintenance, and operations with the critical evaluation of code and design decisions produced by a large language model (LLM). You will work in groups of 4–5 students for 7 weeks. Each group will have a small system consisting of a simulated legacy device, an edge gateway, and a cloud service. The initial system will be functional but intentionally incomplete: it may have limited test coverage, an unfinished CI/CD pipeline, insufficient monitoring, and an outdated key-establishment mechanism. The goal is not to build a large production system or to implement ML-KEM from scratch. Instead, you will use an existing, suitable cryptographic library and focus on the safe integration of ML-KEM into a legacy software environment.

LLM evaluation and reporting. The LLM must be treated as an assistant and as an untrusted source of code and design suggestions, not as an authority. Your group must critically evaluate the generated artefacts by reviewing their correctness, security, maintainability, testability, performance, and operational suitability. Important outputs must be verified through appropriate tests, code review, static analysis, documentation, and, where relevant, failure and security testing. The group must report the prompts used for significant tasks, the relevant LLM outputs or a representative selection of them, the context provided to the LLM, and the changes made by the students. For each generated artifact, explain what was accepted, modified, or rejected; why these decisions were made; how potential problems were identified; and what risks or uncertainties remain. The purpose is not to demonstrate that the LLM produced perfect code, but to demonstrate that the group can assess, validate, improve, and take responsibility for AI-assisted software development.

2 Project objectives

During the project, your group will use an LLM to generate or substantially design selected software artifacts, critically review and evaluate the generated code and proposed solutions, identify functional, security, maintainability, and operational problems, and improve the legacy system through testing, refactoring, and integration. You will integrate ML-KEM into at least one relevant communication path, design and document a realistic migration strategy for legacy devices, build and improve a CI/CD pipeline, and deploy the system to a test environment. In addition, you will add basic monitoring, logging, and post-quantum-related observability, while documenting your technical decisions, the limitations of your solution, and any remaining risks. Possible LLM-generated artefacts include gateway functionality, automated tests, Docker configuration, CI/CD pipeline definitions, deployment configuration, monitoring components, or an initial ML-KEM integration proposal. The output generated must be treated as untrusted. A solution that runs successfully is not necessarily correct, secure, maintainable, or suitable for operations.

3 How to complete the work

Your project should include the following:

- AI-assisted development: Generate selected components with an LLM and record the relevant prompts, outputs, and decisions.
- Baseline analysis: Understand and document the initial architecture, run the system, and establish baseline tests and measurements.
- Critical evaluation and maintenance: Test, review, correct, and improve the generated artefacts; document how problems were discovered and verified.
- ML-KEM integration: Integrate ML-KEM using an existing implementation. Preserve legacy compatibility where appropriate or provide a clearly justified migration and fallback strategy.
- Operations: Automate build and testing, implement deployment to a test environment, and add health checks, logs, and meaningful metrics.
- Final evaluation: Analyze and demonstrate its strengths and weaknesses and identify remaining technical debt and risks.

4 Final submission

Your final submission will consist of

- Part 1: Technical documentation, source code, CI/CD and deployment configuration

- Part 2: Project report (max. 5 pages). The report should examine the project from one (choose only one) of the following three perspectives:
 - 1) Cryptography and mathematics
 - 2) Quantum threats and attackers
 - 3) Information security and organizational cybersecurity
- Part 3: Individual reflection (400-500 words). Each student must submit an individual reflection describing how the project affected their understanding and perspective on AI-assisted software development, legacy modernization, post-quantum cryptography, and software operations. The reflection should discuss what new concepts and skills the student learned, what remains unclear, which parts of the project were difficult or easy and why, and how the project related to the student's previous studies, professional experience, or technical background. It should also consider the student's own contribution and collaboration within the group, as well as any assumptions or opinions that changed during the project. Finally, describe whether the project created interest in further study, experimentation, or professional work related to LLMs, software maintenance, cybersecurity, or post-quantum cryptography. The reflection should be analytical rather than merely a description of completed tasks.

5 Schedule

- Project starts: Mon 7th Sep
- Project final DL: Sun 25th Oct
- DL for the project submission for the peer reviews: Sun 11th Oct
- DL for the peer reviews to be ready: Wed 14th Oct
- Presentations: Thu and Fri, 15th and 16th Oct
- Q&A sessions hybrid or online on every Tuesday 15.9. - 13.10.2026

6 Grading

Overall the project is 60 % of the course points. The 100 % of the project points will consist of

- Part 1: 40 %
- Part 2: 40 %
- Part 3: 10 %
- Peer review and presentation: 10 %

7 Contact

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